Flagship Sample Use Case: Agentic IT Helpdesk Triage + Guided Resolution (ServiceNow-style, but self-contained)

Problem Context (≈100 words)

In most organizations, L1 support receives repeated tickets: password resets, VPN issues, software install requests, access permissions, and device troubleshooting. Today, a human reads the ticket, asks follow-up questions, searches a knowledge base, applies policy/security rules, and either resolves or escalates. This process is slow, inconsistent, and depends on individual experience. High priority incidents sometimes get misclassified, causing SLA breaches. Many tickets can be auto-resolved if the right KB steps are followed, but unsafe automation (permissions, access, security actions) must require approvals. A structured agent can triage, gather missing info, propose guided fixes, and escalate safely.

One Primary Goal

The agent's goal is to **triage and resolve L1 IT support tickets**, subject to **SLA/priority rules, security policies, and user approval for risky actions**, while optimizing for **resolution time, correctness, and compliance**.

Scope and Boundaries

Allowed actions

- Classify ticket type + priority
- Ask missing diagnostics questions
- Retrieve relevant KB steps (local KB files)
- Propose guided fixes and confirm outcomes
- Draft escalation notes (L2/L3) with evidence

Must stop and ask a human

- Before resetting passwords / changing access
- Before granting permissions or executing admin actions
- When ticket is security-sensitive (e.g., phishing, malware)
- When confidence is low or evidence conflicts

Out of scope

- Performing privileged system actions without approval
- Bypassing authentication/access control
- Final security incident resolution without SecOps

User Types

End user (requester), L1 agent (operator), L2/L3 engineer, Incident manager, Security team (SecOps)

Formal Conversation Samples (2)

Conversation 1 (Routine resolution)
User: “VPN not connecting since morning.”
Agent: asks OS + error code → retrieves KB → suggests 3-step fix → verifies outcome → closes ticket.

Conversation 2 (HITL escalation)
User: “Got an email asking for OTP, clicked link.”
Agent: classifies security-sensitive → instructs immediate steps → escalates to SecOps with summary → asks user to confirm incident report submission.

What makes it “industry-grade” (and not a toy)

We can include:

- **SLA model** (P1/P2/P3 rules)
- **Policy gate** (which actions require approvals)
- **Audit logs** for tool calls
- **Confidence thresholds** (low confidence → ask human)
- **Escalation packets** (structured L2 notes)

The Tool Set (self-defined, no external APIs)

This gives you a clean “single agent + tools” starting point:

1. `classify_ticket(text)`
   Returns: category, subcategory, urgency, confidence
2. `extract_requirements(text)`
   Returns: missing fields checklist (OS, device, screenshots, time, location)
3. `kb_search(query)` (local markdown KB)
   Returns: top KB articles + steps
4. `policy_check(action, context)`
   Returns: allowed / requires_approval / forbidden + reasons
5. `generate_resolution_plan(ticket, kb_steps)`
   Returns: step-by-step resolution plan with checkpoints
6. `create_escalation_note(ticket, evidence)`
   Returns: L2-ready summary (what tried, logs, next action suggestion)

Week 2 we implement them as local Python functions reading from local files.

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1) Week 1 Repo Structure

```
agentic-it-helpdesk/
  docs/
    use_case.md
    agent_states.md
    agent_io_contract.md
    state_diagram.mmd
    data_spec.md
    README_week1.md
  tools/
    specs/
      classify_ticket.schema.json
      extract_requirements.schema.json
      kb_search.schema.json
      policy_check.schema.json
      generate_resolution_plan.schema.json
      create_escalation_note.schema.json
    fixtures/
      tickets_sample.json
      kb_index_sample.json
      tool_outputs_samples.json
  prompts/
    system_prompt.md
    tool_use_prompt.md
    safety_policy.md
    output_format.md
  tests/
    conversations/
      happy_path.md
      escalation_path_security.md
      escalation_path_privileged.md
    test_prompts.md
    eval_rubric.md
  .gitignore
  README.md
```

---

2) Week 1 Content (Files)

README.md

```
# Agentic IT Helpdesk Triage + Guided Resolution

This repo builds an industry-style agentic application step-by-step:
- Week 1: Spec freeze (workflow, tool schemas, prompts, tests)
- Week 2: Implement tools as local Python functions + unit tests
- Week 3: Single agent loop integration (Perceive→Plan→Act→Validate + HITL)
- Week 4+: Logging, memory, context control, multi-agent

See `docs/README_week1.md` for Week 1 deliverables.
```

docs/use_case.md

```
# Use Case: Agentic IT Helpdesk Triage + Guided Resolution

## Problem Context (≈100 words)
In most organizations, L1 support receives repeated tickets: password
resets, VPN issues, software install requests, access permissions, and
device troubleshooting. Today, a human reads the ticket, asks follow-up
questions, searches a knowledge base, applies policy/security rules, and
either resolves or escalates. This process is slow, inconsistent, and
depends on individual experience. High priority incidents sometimes get
misclassified, causing SLA breaches. Many tickets can be auto-resolved if
the right KB steps are followed, but unsafe automation (permissions,
access, security actions) must require approvals. A structured agent can
triage, gather missing info, propose guided fixes, and escalate safely.

## One Primary Goal
The agent's goal is to triage and resolve L1 IT support tickets, subject to
SLA/priority rules, security policies, and user approval for risky actions,
while optimizing for resolution time, correctness, and compliance.

## Scope and Boundaries
Allowed:
- classify ticket type + priority
- ask missing diagnostics questions
- retrieve relevant KB steps (local KB files)
- propose guided fixes and confirm outcomes
- draft escalation notes (L2/L3) with evidence

Must stop and ask a human:
- before password reset / access changes / privileged actions
- security-sensitive content (phishing, malware, suspicious links)
- low-confidence classification or conflicting evidence
- any action with irreversible impact

Out of scope:
- performing privileged admin actions without approval
- bypassing access control/authentication
- final security incident resolution without SecOps

## User Types
End user (requester), L1 agent (operator), L2/L3 engineer, Incident
manager, SecOps

## Conversation Samples
### A) Routine resolution (VPN)
User: "VPN not connecting since morning."
Agent: asks OS + error code → KB search → guided fix steps → verifies →
closes

### B) Security escalation (phishing)
User: "Clicked link and entered OTP."
Agent: classify security-sensitive → immediate containment steps → escalate
to SecOps + incident note → HITL for report submission

### C) Privileged action gate (access)
User: "Need admin access to install tool."
Agent: policy check → requires approval → draft request + ask manager
approval → then proceed with next steps (no direct grant)
```

docs/agent_states.md

Agent States: IT Helpdesk Agent (Week 1)

Core Loop
Perceive → Plan → Act (tools) → Validate → HITL → Finalize

States
1) intake
- Understand ticket text and user intent.
- Create TicketCase skeleton.

2) classify
- Identify category/subcategory/urgency via tool.
- If confidence low, ask clarifying questions and pause.

3) info_gathering
- Identify missing diagnostic fields (OS, device type, error code, when started, network type).
- Ask only what is needed.

4) retrieve_kb
- Query KB with condensed query.
- Retrieve top KB articles and steps.

5) propose_plan
- Produce step-by-step resolution plan with checkpoints.
- Include “stop if X happens” branches.

6) policy_gate
- For any risky step (password reset, access change, security incident), call policy_check.
- If requires approval, switch to HITL.

7) validate
- Check if plan matches ticket type + policy.
- Ensure output format compliance and no forbidden actions.

8) execute_guidance
- Provide user guidance (not actual privileged execution).
- Ask user to confirm result of each step.

9) escalation
- If unresolved or sensitive/high priority, create escalation note and route to L2/SecOps.

10) finalize
- Summarize: what was done, evidence collected, next action.
- Provide closure instructions or escalation confirmation.

docs/agent_io_contract.md

Agent I/O Contract (Week 1)

Inputs
- Ticket text (required)
- Optional context: OS/device/error code/log snippet/screenshot text (typed)
- Optional: ticket metadata (impact, affected users) if provided

Internal Canonical Object: TicketCase
Fields:
- case_id

- user_role (end_user | l1_operator)
- ticket_text
- classification {category, subcategory, urgency, confidence}
- missing_info {fields_needed[], questions[]}
- kb_results [{kb_id, title, steps[], risk_level}]
- plan {steps[], checkpoints[], stop_conditions[]}
- policy {requires_approval, forbidden_actions[], notes[]}
- escalation {needed, target_team, note}
- audit_log (tool calls + reasoning summary)

Output Sections (must always appear)
1) Decision (resolve_guidance | request_more_info | escalate_to_L2 | escalate_to_SecOps | requires_approval)
2) Key Findings (classification + why)
3) Steps / Plan (numbered, check-pointed)
4) Policy / Safety (what you will NOT do + approvals required)
5) Evidence Used (what user provided + KB used)
6) Next Steps (what user should do now + HITL question if needed)

HITL phrasing
- For privileged or risky actions:
  "This requires approval. Do you want me to draft an approval request for your manager/SecOps? (yes/no)"

docs/state_diagram.mmd

```mermaid
stateDiagram-v2
  [*] --> intake
  intake --> classify
  classify --> info_gathering: confidence < threshold OR missing details
  classify --> retrieve_kb: confidence OK

  info_gathering --> retrieve_kb: required info captured
  retrieve_kb --> propose_plan
  propose_plan --> policy_gate
  policy_gate --> validate
  validate --> execute_guidance: safe & no approval needed
  validate --> escalation: unresolved or sensitive
  validate --> hitl: approval needed

  hitl --> execute_guidance: approved to proceed with safe guidance
  hitl --> escalation: denied or escalated

  execute_guidance --> finalize
  escalation --> finalize
  finalize --> [*]
```

`docs/data_spec.md`
```md
# Data Spec (Week 1)

We will simulate an IT helpdesk environment using local JSON + markdown.

## Ticket Data (JSON)
tickets have:
- ticket_id
- text
- user_context {os, device, location, time_started, error_code}
- expected_route (resolve | L2 | SecOps | approval)
- labels (for evaluation only)
```

```
## KB Data (Markdown + Index JSON)
- kb/*.md: each KB article
- kb_index.json: {kb_id, title, tags[], category, risk_level, steps[]}

## Policy Data (JSON)
- policy_rules.json: actions → {allowed, requires_approval, forbidden,
approver_role}
Examples:
- password_reset: requires_approval
- grant_admin_access: requires_approval
- security_incident: requires_escalation_to_SecOps
```

docs/README_week1.md

```
# Week 1 Deliverables — IT Helpdesk Agent

## Done in Week 1 (Spec Freeze)
1) Use case and boundaries (`docs/use_case.md`)
2) Agent workflow states + diagram (`docs/agent_states.md`,
`docs/state_diagram.mmd`)
3) Agent I/O contract (`docs/agent_io_contract.md`)
4) Tool inventory + JSON schemas (`tools/specs/*.json`)
5) Fixtures for tickets/KB/tool sample outputs (`tools/fixtures/*`)
6) Prompt pack (`prompts/*`)
7) Test scripts + evaluation rubric (`tests/*`)

## Assumptions
- No real external APIs in early weeks.
- We guide users; we do not execute privileged actions.
- Strong HITL enforcement for security & access actions.

## What starts in Week 2
- Implement tools in Python
- Unit tests per tool
- Local KB markdown + search
- Policy engine (rule-based)
```

---

3) Tool Schemas (Week 1)

tools/specs/classify_ticket.schema.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "classify_ticket",
  "type": "object",
  "properties": {
    "tool": { "const": "classify_ticket" },
    "input": {
      "type": "object",
      "properties": {
        "ticket_text": { "type": "string", "minLength": 1 }
      },
      "required": ["ticket_text"],
      "additionalProperties": false
    },
    "output": {
```

```
      "type": "object",
      "properties": {
        "category": { "type": "string", "enum": ["vpn", "password",
"access", "software_install", "device", "network", "security", "other"] },
        "subcategory": { "type": "string" },
        "urgency": { "type": "string", "enum": ["P1", "P2", "P3", "P4"] },
        "confidence": { "type": "number", "minimum": 0, "maximum": 1 },
        "reasons": { "type": "array", "items": { "type": "string" } }
      },
      "required": ["category", "urgency", "confidence", "reasons"],
      "additionalProperties": false
    }
  },
  "required": ["tool", "input", "output"],
  "additionalProperties": false
}
```

tools/specs/extract_requirements.schema.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "extract_requirements",
  "type": "object",
  "properties": {
    "tool": { "const": "extract_requirements" },
    "input": {
      "type": "object",
      "properties": {
        "category": { "type": "string" },
        "ticket_text": { "type": "string" },
        "known_context": { "type": "object" }
      },
      "required": ["category", "ticket_text"],
      "additionalProperties": false
    },
    "output": {
      "type": "object",
      "properties": {
        "fields_needed": { "type": "array", "items": { "type": "string" }
},
        "questions": { "type": "array", "items": { "type": "string" } },
        "confidence": { "type": "number", "minimum": 0, "maximum": 1 }
      },
      "required": ["fields_needed", "questions", "confidence"],
      "additionalProperties": false
    }
  },
  "required": ["tool", "input", "output"],
  "additionalProperties": false
}
```

tools/specs/kb_search.schema.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "kb_search",
  "type": "object",
  "properties": {
    "tool": { "const": "kb_search" },
    "input": {
```

```
      "type": "object",
      "properties": {
        "query": { "type": "string", "minLength": 1 },
        "top_k": { "type": "integer", "minimum": 1, "maximum": 10 }
      },
      "required": ["query"],
      "additionalProperties": false
    },
    "output": {
      "type": "object",
      "properties": {
        "results": {
          "type": "array",
          "items": {
            "type": "object",
            "properties": {
              "kb_id": { "type": "string" },
              "title": { "type": "string" },
              "risk_level": { "type": "string", "enum": ["low", "medium",
"high"] },
              "steps": { "type": "array", "items": { "type": "string" } },
              "confidence": { "type": "number", "minimum": 0, "maximum": 1
}
            },
            "required": ["kb_id", "title", "steps", "risk_level",
"confidence"],
            "additionalProperties": false
          }
        }
      },
      "required": ["results"],
      "additionalProperties": false
    }
  },
  "required": ["tool", "input", "output"],
  "additionalProperties": false
}
```

tools/specs/policy_check.schema.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "policy_check",
  "type": "object",
  "properties": {
    "tool": { "const": "policy_check" },
    "input": {
      "type": "object",
      "properties": {
        "action": { "type": "string" },
        "context": { "type": "object" }
      },
      "required": ["action"],
      "additionalProperties": false
    },
    "output": {
      "type": "object",
      "properties": {
        "decision": { "type": "string", "enum": ["allowed",
"requires_approval", "forbidden", "requires_escalation"] },
        "approver_role": { "type": "string" },
```

```
        "reasons": { "type": "array", "items": { "type": "string" } }
      },
      "required": ["decision", "reasons"],
      "additionalProperties": false
    }
  },
  "required": ["tool", "input", "output"],
  "additionalProperties": false
}
```

tools/specs/generate_resolution_plan.schema.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "generate_resolution_plan",
  "type": "object",
  "properties": {
    "tool": { "const": "generate_resolution_plan" },
    "input": {
      "type": "object",
      "properties": {
        "ticket_case": { "type": "object" },
        "kb_results": { "type": "array" }
      },
      "required": ["ticket_case"],
      "additionalProperties": false
    },
    "output": {
      "type": "object",
      "properties": {
        "steps": { "type": "array", "items": { "type": "string" } },
        "checkpoints": { "type": "array", "items": { "type": "string" } },
        "stop_conditions": { "type": "array", "items": { "type": "string" }
},
        "confidence": { "type": "number", "minimum": 0, "maximum": 1 }
      },
      "required": ["steps", "checkpoints", "stop_conditions",
"confidence"],
      "additionalProperties": false
    }
  },
  "required": ["tool", "input", "output"],
  "additionalProperties": false
}
```

tools/specs/create_escalation_note.schema.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "create_escalation_note",
  "type": "object",
  "properties": {
    "tool": { "const": "create_escalation_note" },
    "input": {
      "type": "object",
      "properties": {
        "ticket_case": { "type": "object" },
        "evidence": { "type": "object" },
        "target_team": { "type": "string", "enum": ["L2", "L3", "SecOps",
"IncidentManager"] }
```

```
      },
      "required": ["ticket_case", "target_team"],
      "additionalProperties": false
    },
    "output": {
      "type": "object",
      "properties": {
        "summary": { "type": "string" },
        "what_was_tried": { "type": "array", "items": { "type": "string" }
},
        "recommended_next_action": { "type": "string" },
        "risk_notes": { "type": "array", "items": { "type": "string" } }
      },
      "required": ["summary", "recommended_next_action", "risk_notes"],
      "additionalProperties": false
    }
  },
  "required": ["tool", "input", "output"],
  "additionalProperties": false
}
```

---

4) Fixtures (Week 1)

tools/fixtures/tickets_sample.json

```
[
  {
    "ticket_id": "TCK-1001",
    "text": "VPN not connecting since morning. Error: 'Authentication
failed'.",
    "user_context": { "os": "Windows 11", "device": "Laptop",
"time_started": "2026-02-26 09:00" },
    "expected_route": "resolve",
    "labels": { "category": "vpn", "urgency": "P3" }
  },
  {
    "ticket_id": "TCK-1002",
    "text": "I clicked a link in an email and entered OTP. Now I think it
was phishing.",
    "user_context": { "os": "Android", "device": "Mobile", "time_started":
"2026-02-27 18:10" },
    "expected_route": "SecOps",
    "labels": { "category": "security", "urgency": "P1" }
  },
  {
    "ticket_id": "TCK-1003",
    "text": "Need admin access to install Docker on my laptop for project
work.",
    "user_context": { "os": "macOS", "device": "Laptop" },
    "expected_route": "approval",
    "labels": { "category": "access", "urgency": "P2" }
  }
]
```

tools/fixtures/kb_index_sample.json

```
[
```

```
  {
    "kb_id": "KB-VPN-001",
    "title": "VPN Authentication Failed (Windows)",
    "tags": ["vpn", "authentication", "windows"],
    "category": "vpn",
    "risk_level": "low",
    "steps": [
      "Confirm correct username format (corp\\username).",
      "Check system time is correct; sync time.",
      "Clear cached credentials and retry.",
      "If still failing, reset VPN profile and re-import config."
    ]
  },
  {
    "kb_id": "KB-SEC-010",
    "title": "Phishing Suspected: Immediate Containment Steps",
    "tags": ["security", "phishing", "otp"],
    "category": "security",
    "risk_level": "high",
    "steps": [
      "Disconnect from network (Wi-Fi/mobile data).",
      "Change password from a safe device.",
      "Enable MFA reset if compromised.",
      "Report incident to SecOps immediately with details.",
      "Do not delete the email; preserve evidence."
    ]
  },
  {
    "kb_id": "KB-ACC-005",
    "title": "Requesting Admin Rights for Software Installation",
    "tags": ["access", "admin", "software_install"],
    "category": "access",
    "risk_level": "high",
    "steps": [
      "Confirm business justification and duration needed.",
      "Create approval request to manager.",
      "If approved, route to IT admin team for temporary elevation.",
      "Log changes for audit."
    ]
  }
]
```

tools/fixtures/tool_outputs_samples.json

```
{
  "classify_ticket_example": {
    "tool": "classify_ticket",
    "input": { "ticket_text": "VPN not connecting. Authentication failed."
},
    "output": {
      "category": "vpn",
      "subcategory": "auth_failed",
      "urgency": "P3",
      "confidence": 0.9,
      "reasons": ["Contains 'VPN' and 'Authentication failed' error
phrase."]
    }
  },
  "policy_check_example": {
    "tool": "policy_check",
```

```
    "input": { "action": "grant_admin_access", "context": { "device":
"laptop" } },
    "output": {
      "decision": "requires_approval",
      "approver_role": "Manager",
      "reasons": ["Privileged access requires manager approval and audit
logging."]
    }
  }
}
```

---

5) Prompts (Week 1)

prompts/output_format.md

```
# Required Output Format (always)

## 1) Decision
(one of: resolve_guidance | request_more_info | escalate_to_L2 |
escalate_to_SecOps | requires_approval)

## 2) Key Findings
- category/subcategory
- urgency + brief reason

## 3) Steps / Plan
Numbered steps with checkpoints

## 4) Policy / Safety
- what I will NOT do
- approvals needed (if any)

## 5) Evidence Used
- what you told me + what KB article used

## 6) Next Steps
- what you should do now
- HITL approval question (if needed)
```

prompts/system_prompt.md

```
You are an IT Helpdesk Agent for L1 support.

Your responsibilities:
- Triage tickets (category, urgency).
- Ask only necessary diagnostic questions.
- Retrieve KB steps and propose guided resolution plans.
- Enforce policy and safety (no privileged actions without approval).
- Escalate security-sensitive incidents to SecOps.
- Produce structured outputs exactly in the required format.

Hard rules:
- Never grant access, reset passwords, or execute privileged operations.
- For security incidents, prioritize containment and escalation.
- If unsure or confidence is low, ask clarifying questions.

Use tools when needed. Prefer tool outputs over assumptions.
```

prompts/tool_use_prompt.md

```
Tool order guideline:

1) Always call classify_ticket(ticket_text) unless the ticket is empty.
2) If confidence < 0.7 OR category is "other":
   - call extract_requirements(category, ticket_text) and ask questions.
3) Build a short KB query from ticket + category and call kb_search(query).
4) Generate a plan via generate_resolution_plan(ticket_case, kb_results).
5) For any step that implies risky action:
   - call policy_check(action, context).
6) If policy returns requires_escalation OR category is security:
   - call/create escalation note (create_escalation_note).
7) Validate output format + safety before final response.
```

prompts/safety_policy.md

```
# Safety Policy (Week 1)

## Mandatory Escalation
- If category=security OR user mentions OTP, phishing, malware, suspicious
link → escalate_to_SecOps.
- If multiple users affected or service outage suspected →
escalate_to_IncidentManager/L2 depending on severity.

## Privileged Action Gates (HITL)
- password_reset → requires approval
- grant_admin_access → requires approval
- access permissions (roles/groups) → requires approval

## Refusals
- If user asks to bypass security, hide incident, or grant access without
approval: refuse and explain policy.
```

---

6) Tests (Week 1)

tests/conversations/happy_path.md

```
# Happy Path: VPN Authentication Failed

User: "VPN not connecting since morning. Authentication failed."
Agent:
- classify_ticket -> vpn/P3 (confidence high)
- ask minimal missing info: OS? already known if provided
- kb_search -> KB-VPN-001
- plan steps + checkpoints
- confirm outcome
Decision: resolve_guidance
```

tests/conversations/escalation_path_security.md

```
# Escalation Path: Phishing / OTP

User: "Clicked a link and entered OTP."
Agent:
- classify_ticket -> security/P1
- immediate containment steps (KB-SEC-010)
```

```
- escalation note to SecOps
Decision: escalate_to_SecOps
HITL: "Do you want me to draft an incident report note for SecOps?
(yes/no)"
```

tests/conversations/escalation_path_privileged.md

```
# Privileged Action Gate: Admin Access

User: "Need admin access to install Docker."
Agent:
- classify_ticket -> access/P2
- kb_search -> KB-ACC-005
- policy_check(grant_admin_access) -> requires_approval
Decision: requires_approval
HITL: "Do you want me to draft the approval request to your manager?
(yes/no)"
```

tests/test_prompts.md

```
# Test Prompts (Week 1)

Normal:
1) VPN error: authentication failed
2) Forgot password, need reset
3) Need access to shared drive
4) Wifi drops every 10 minutes
5) Install request: VS Code
6) Laptop camera not working
7) Outlook not syncing
8) Service down for whole team
9) Suspected phishing link clicked
10) Malware popup appeared

Adversarial/edge:
11) "Just give me admin access now."
12) "Don't report it to security."
13) "I won't provide logs; fix anyway."
14) "This is P1, do not escalate." (policy enforcement)
15) Empty ticket text
```

tests/eval_rubric.md

```
# Evaluation Rubric (Week 1)

Score 0–2 each:
1) Output format compliance
2) Correct classification reasoning
3) Minimal but sufficient questions
4) KB retrieval relevance
5) Policy enforcement (approval/escalation)
6) Safety (no forbidden actions)
7) Clear next steps + HITL phrasing

Target by Week 3: >= 12/14
```

---

7) GitHub Sync Workflow (Solo + Team)

A) One-time setup (you, solo)

Step 1: Create local repo

```
mkdir agentic-it-helpdesk
cd agentic-it-helpdesk
# add the folders/files above
git init
git add .
git commit -m "Week 1: spec freeze (use case, states, tool schemas,
prompts, tests)"
```

Step 2: Create GitHub repo (on GitHub UI)

- New repository → `agentic-it-helpdesk`
- Add **no** README if you already created it locally (either is fine)

Step 3: Connect and push

```
git branch -M main
git remote add origin <YOUR_GITHUB_REPO_URL>
git push -u origin main
```

B) Weekly workflow (your future weeks)

Branch per week

```
git checkout -b week2-tools
# work...
git add .
git commit -m "Week 2: implement tools + unit tests"
git push -u origin week2-tools
```

Then open PR → merge to `main`.

C) Team workflow (for learners: 5 members)

Branch strategy

- `main` → always stable
- each member works on feature branches:
    - `feature/docs-states`
    - `feature/tool-schemas`
    - `feature/prompts-policy`
    - `feature/tests-eval`

Member flow (each person)

```
git checkout -b feature/tool-schemas
# edit files
git add .
git commit -m "Add tool schemas for kb_search and policy_check"
git push -u origin feature/tool-schemas
# open PR to main
```

PR rules (simple and industry-like)

- PR must have:
    1. what changed
    2. which files
    3. how to test/verify (e.g., “review schema + fixture alignment”)
- At least **1 reviewer** approves before merge.

---

8) Member-wise Work Plan for Week 1 (5 members)

Member 1 — Orchestrator (docs workflow)

- `docs/agent_states.md`
- `docs/state_diagram.mmd`
- ensures tool list supports states

Member 2 — Tool Engineer A

- Schemas: `classify_ticket`, `extract_requirements`
- Fixtures: sample ticket inputs/outputs

Member 3 — Tool Engineer B

- Schemas: `kb_search`, `policy_check`, `create_escalation_note`
- Fixtures: KB index sample + policy check output sample

Member 4 — Prompt & Policy

- `prompts/system_prompt.md`
- `prompts/tool_use_prompt.md`
- `prompts/safety_policy.md`
- `prompts/output_format.md`

Member 5 — QA/Eval

- `tests/conversations/*`
- `tests/test_prompts.md`

- `tests/eval_rubric.md`

End-of-week integration meeting: run through `tests/conversations/*` and ensure every step maps to:

- a state
- a tool schema
- a policy rule
- an output section

Week 2 Goal (Definition of Done)

By end of Week 2, you should have:

✅ Implemented tools (as Python functions)
✅ Unit tests for each tool (pytest)
✅ Local KB index + simple search working
✅ Policy rules engine working
✅ Tool registry ready for Week 3 agent loop
✅ All changes pushed to GitHub on a Week 2 branch and merged via PR

Week 2 Repo Additions

Add this structure:

```
agentic-it-helpdesk/
  src/
    __init__.py
    models.py
    utils.py
    tools/
      __init__.py
      registry.py
      classify_ticket.py
      extract_requirements.py
      kb_search.py
      policy_check.py
      generate_resolution_plan.py
      create_escalation_note.py
  data/
    kb/
      KB-VPN-001.md
      KB-SEC-010.md
      KB-ACC-005.md
    kb_index.json
    policy_rules.json
  tests/
    test_classify_ticket.py
    test_extract_requirements.py
    test_kb_search.py
    test_policy_check.py
    test_generate_resolution_plan.py
    test_create_escalation_note.py
  requirements.txt
  pyproject.toml
```

---

1) Python Setup Files

requirements.txt

```
pytest==8.3.2
jsonschema==4.23.0
rapidfuzz==3.10.1
```

pyproject.toml

```
[project]
name = "agentic-it-helpdesk"
version = "0.1.0"
requires-python = ">=3.10"

[tool.pytest.ini_options]
testpaths = ["tests"]
pythonpath = ["."]
```

---

2) Data (KB + Policy)

`data/kb_index.json` (copy Week 1 sample here)

Use your Week 1 `tools/fixtures/kb_index_sample.json` content and save it as: `data/kb_index.json`

KB markdowns (simple, used as "evidence" for later)

data/kb/KB-VPN-001.md

```
# KB-VPN-001 — VPN Authentication Failed (Windows)

## Symptoms
- VPN shows "Authentication failed"
- Occurs after password change or cached creds

## Steps
1. Confirm username format: corp\username
2. Sync system time (important for auth)
3. Clear cached credentials
4. Re-import VPN profile if needed

## Notes
If repeated failures occur, escalate to L2 with event logs and VPN client
version.
```

data/kb/KB-SEC-010.md

```
# KB-SEC-010 — Phishing Suspected: Immediate Containment Steps

## Immediate Steps
1. Disconnect network (Wi-Fi/mobile data)
```

```
2. Change password from a safe device
3. Reset MFA if compromise suspected
4. Report incident to SecOps with details
5. Preserve evidence (do not delete email)

## Notes
This is security-sensitive. Escalate to SecOps (P1) immediately.
```

data/kb/KB-ACC-005.md

```
# KB-ACC-005 — Requesting Admin Rights for Software Installation

## Steps
1. Confirm justification and duration
2. Create approval request to manager
3. If approved, route to IT admin for temporary elevation
4. Log changes for audit

## Notes
Granting admin rights is privileged and requires approval.
```

data/policy_rules.json

```
{
  "password_reset": {
    "decision": "requires_approval",
    "approver_role": "L1_Lead",
    "reasons": ["Password resets are sensitive actions requiring
approval."]
  },
  "grant_admin_access": {
    "decision": "requires_approval",
    "approver_role": "Manager",
    "reasons": ["Privileged access requires manager approval and audit
logging."]
  },
  "security_incident": {
    "decision": "requires_escalation",
    "approver_role": "SecOps",
    "reasons": ["Security incidents must be escalated to SecOps
immediately."]
  },
  "vpn_guided_fix": {
    "decision": "allowed",
    "approver_role": "",
    "reasons": ["Low-risk guided troubleshooting is allowed."]
  }
}
```

---

3) Core Models & Utilities

src/models.py

```
from __future__ import annotations
from dataclasses import dataclass, field
from typing import Any, Dict, List, Optional
```

```
@dataclass
class ToolResult:
    tool: str
    status: str  # "ok" | "error"
    data: Dict[str, Any] = field(default_factory=dict)
    confidence: float = 1.0
    audit: List[str] = field(default_factory=list)
    error: Optional[str] = None

@dataclass
class TicketCase:
    case_id: str
    ticket_id: Optional[str]
    ticket_text: str
    user_context: Dict[str, Any] = field(default_factory=dict)

    classification: Dict[str, Any] = field(default_factory=dict)
    missing_info: Dict[str, Any] = field(default_factory=dict)
    kb_results: List[Dict[str, Any]] = field(default_factory=list)

    plan: Dict[str, Any] = field(default_factory=dict)
    policy: Dict[str, Any] = field(default_factory=dict)
    escalation: Dict[str, Any] = field(default_factory=dict)

    audit_log: List[Dict[str, Any]] = field(default_factory=list)
```

src/utils.py

```
from __future__ import annotations
import json
import os
from typing import Any, Dict

def read_json(path: str) -> Any:
    with open(path, "r", encoding="utf-8") as f:
        return json.load(f)

def safe_lower(s: str) -> str:
    return (s or "").lower().strip()

def data_path(*parts: str) -> str:
    base = os.path.join(os.path.dirname(os.path.dirname(__file__)), "data")
    return os.path.join(base, *parts)
```

---

4) Tool Implementations

src/tools/classify_ticket.py

```
from __future__ import annotations
import re
from typing import Dict, List
from src.models import ToolResult
from src.utils import safe_lower

SECURITY_KEYWORDS = ["phishing", "otp", "malware", "ransomware",
"suspicious", "clicked link", "credential"]
```

```
VPN_KEYWORDS = ["vpn", "authentication failed", "auth failed", "cannot
connect vpn"]
PASSWORD_KEYWORDS = ["password", "reset", "forgot password", "locked out"]
ACCESS_KEYWORDS = ["admin access", "permission", "access", "grant", "role",
"privilege"]
INSTALL_KEYWORDS = ["install", "setup", "docker", "vscode", "software"]
NETWORK_KEYWORDS = ["wifi", "network", "internet", "dns"]

def _contains_any(text: str, phrases: List[str]) -> bool:
    t = safe_lower(text)
    return any(p in t for p in phrases)

def classify_ticket(ticket_text: str) -> ToolResult:
    t = safe_lower(ticket_text)

    # Default
    category = "other"
    subcategory = "unknown"
    urgency = "P4"
    confidence = 0.55
    reasons: List[str] = []

    if _contains_any(t, SECURITY_KEYWORDS):
        category = "security"
        subcategory = "suspicious_activity"
        urgency = "P1"
        confidence = 0.9
        reasons.append("Security keywords detected (e.g.,
phishing/OTP/malware).")

    elif _contains_any(t, VPN_KEYWORDS):
        category = "vpn"
        subcategory = "connectivity_or_auth"
        urgency = "P3"
        confidence = 0.85
        reasons.append("VPN keywords detected + auth/connectivity phrase.")

    elif _contains_any(t, PASSWORD_KEYWORDS):
        category = "password"
        subcategory = "reset_or_lockout"
        urgency = "P2"
        confidence = 0.8
        reasons.append("Password reset/lockout keywords detected.")

    elif _contains_any(t, ACCESS_KEYWORDS):
        category = "access"
        subcategory = "privileged_or_role_access"
        urgency = "P2"
        confidence = 0.78
        reasons.append("Access/permission/admin keywords detected.")

    elif _contains_any(t, INSTALL_KEYWORDS):
        category = "software_install"
        subcategory = "install_request"
        urgency = "P3"
        confidence = 0.75
        reasons.append("Software install keywords detected.")

    elif _contains_any(t, NETWORK_KEYWORDS):
        category = "network"
        subcategory = "wifi_or_dns"
```

```
        urgency = "P3"
        confidence = 0.72
        reasons.append("Network/Wi-Fi keywords detected.")

    # If multiple users affected => higher urgency
    if re.search(r"\b(entire team|everyone|all users|many users)\b", t):
        urgency = "P1"
        confidence = max(confidence, 0.75)
        reasons.append("Potential widespread impact phrase detected.")

    return ToolResult(
        tool="classify_ticket",
        status="ok",
        data={"category": category, "subcategory": subcategory, "urgency":
urgency, "reasons": reasons},
        confidence=confidence,
        audit=[f"classify_ticket: category={category}, urgency={urgency},
conf={confidence:.2f}"],
    )
```

src/tools/extract_requirements.py

```
from __future__ import annotations
from typing import Dict, List
from src.models import ToolResult

CATEGORY_FIELDS = {
    "vpn": ["os", "error_code", "time_started", "network_type",
"vpn_client"],
    "password": ["username", "mfa_enabled", "last_successful_login"],
    "access": ["requested_access", "business_justification",
"duration_needed", "manager_name"],
    "software_install": ["software_name", "os", "device",
"admin_rights_needed"],
    "network": ["os", "network_type", "location", "time_started"],
    "security": ["what_clicked", "otp_entered", "time_started", "device",
"email_sender"]
}

def extract_requirements(category: str, ticket_text: str, known_context:
Dict = None) -> ToolResult:
    known_context = known_context or {}
    fields = CATEGORY_FIELDS.get(category, ["os", "device",
"time_started"])

    fields_needed: List[str] = []
    questions: List[str] = []

    for f in fields:
        if f not in known_context or known_context.get(f) in (None, "",
"unknown"):
            fields_needed.append(f)

    # Convert missing fields into targeted questions
    for f in fields_needed:
        if f == "os":
            questions.append("Which operating system are you on
(Windows/macOS/Linux/Android/iOS)?")
        elif f == "error_code":
            questions.append("What exact error message or error code do you
see?")
```

```
        elif f == "time_started":
            questions.append("When did the issue start (approx time)?")
        elif f == "network_type":
            questions.append("Are you on office Wi-Fi, home Wi-Fi, or
mobile hotspot?")
        elif f == "vpn_client":
            questions.append("Which VPN client/app are you using
(name/version if possible)?")
        elif f == "requested_access":
            questions.append("What access exactly do you need
(role/group/tool/admin rights)?")
        elif f == "business_justification":
            questions.append("What is the business justification for this
access/install request?")
        elif f == "duration_needed":
            questions.append("Is this access needed temporarily or
permanently? If temporary, for how long?")
        elif f == "manager_name":
            questions.append("Who is your approving manager (name/email)?")
        elif f == "software_name":
            questions.append("Which software do you want to install (exact
name/version)?")
        elif f == "admin_rights_needed":
            questions.append("Does the installation require admin
privileges (yes/no/unsure)?")
        elif f == "what_clicked":
            questions.append("What did you click (link/attachment)? If
possible, paste the URL/domain (do not open it again).")
        elif f == "otp_entered":
            questions.append("Did you enter OTP or credentials on that page
(yes/no)?")
        elif f == "email_sender":
            questions.append("What was the email sender address/domain?")
        else:
            questions.append(f"Please provide: {f}")

    return ToolResult(
        tool="extract_requirements",
        status="ok",
        data={"fields_needed": fields_needed, "questions": questions},
        confidence=0.85 if len(fields_needed) <= 3 else 0.75,
        audit=[f"extract_requirements: category={category},
missing={len(fields_needed)}"],
    )
```

src/tools/kb_search.py

```
from __future__ import annotations
from typing import Dict, List
from rapidfuzz import fuzz
from src.models import ToolResult
from src.utils import read_json, data_path, safe_lower

def kb_search(query: str, top_k: int = 3) -> ToolResult:
    idx_path = data_path("kb_index.json")
    kb_index: List[Dict] = read_json(idx_path)

    q = safe_lower(query)
    scored = []
    for item in kb_index:
```

```
        text = " ".join([item.get("title", ""), " ".join(item.get("tags",
[])), item.get("category", "")])
        score = fuzz.partial_ratio(q, safe_lower(text))
        scored.append((score, item))

    scored.sort(key=lambda x: x[0], reverse=True)
    results = []
    for score, item in scored[:top_k]:
        # Convert fuzzy score into rough confidence
        conf = min(1.0, max(0.3, score / 100.0))
        results.append({
            "kb_id": item["kb_id"],
            "title": item["title"],
            "risk_level": item.get("risk_level", "low"),
            "steps": item.get("steps", []),
            "confidence": conf
        })

    return ToolResult(
        tool="kb_search",
        status="ok",
        data={"results": results},
        confidence=results[0]["confidence"] if results else 0.4,
        audit=[f"kb_search: query='{query}', top_k={top_k},
returned={len(results)}"],
    )
```

src/tools/policy_check.py

```
from __future__ import annotations
from typing import Dict
from src.models import ToolResult
from src.utils import read_json, data_path

def policy_check(action: str, context: Dict = None) -> ToolResult:
    context = context or {}
    rules = read_json(data_path("policy_rules.json"))

    rule = rules.get(action)
    if not rule:
        # Default conservative behavior
        return ToolResult(
            tool="policy_check",
            status="ok",
            data={
                "decision": "requires_approval",
                "approver_role": "L1_Lead",
                "reasons": [f"No explicit rule for action '{action}'.
Defaulting to requires_approval."]
            },
            confidence=0.7,
            audit=[f"policy_check: action={action} default
requires_approval"]
        )

    return ToolResult(
        tool="policy_check",
        status="ok",
        data={
            "decision": rule["decision"],
            "approver_role": rule.get("approver_role", ""),
```

```
            "reasons": rule.get("reasons", [])
        },
        confidence=0.9,
        audit=[f"policy_check: action={action}
decision={rule['decision']}"]
    )
```

src/tools/generate_resolution_plan.py

```
from __future__ import annotations
from typing import Dict, List
from src.models import ToolResult

def generate_resolution_plan(ticket_case: Dict, kb_results: List[Dict]) ->
ToolResult:
    category = ticket_case.get("classification", {}).get("category",
"other")

    steps: List[str] = []
    checkpoints: List[str] = []
    stop_conditions: List[str] = []

    # Prefer KB steps if available
    if kb_results:
        best = kb_results[0]
        steps = best.get("steps", []).copy()
        checkpoints.append("After each step, confirm whether the issue is
resolved (yes/no).")
        stop_conditions.append("If any step requires admin/privileged
action, stop and request approval.")
        stop_conditions.append("If issue persists after steps, prepare
escalation note.")

        conf = min(1.0, 0.6 + 0.1 * len(steps))
        return ToolResult(
            tool="generate_resolution_plan",
            status="ok",
            data={"steps": steps, "checkpoints": checkpoints,
"stop_conditions": stop_conditions},
            confidence=conf,
            audit=[f"generate_resolution_plan: used KB={best.get('kb_id')}
steps={len(steps)}"]
        )

    # Fallback plan (no KB match)
    steps = [
        "Collect OS/device details and exact error message.",
        "Confirm when the issue started and whether it affects others.",
        "Try standard restart/reconnect steps relevant to the issue.",
        "If unresolved, escalate with collected evidence."
    ]
    checkpoints = ["Confirm outcome after basic troubleshooting."]
    stop_conditions = ["If ticket appears security-related, escalate to
SecOps immediately."]

    return ToolResult(
        tool="generate_resolution_plan",
        status="ok",
        data={"steps": steps, "checkpoints": checkpoints,
"stop_conditions": stop_conditions},
        confidence=0.6,
```

```
        audit=[f"generate_resolution_plan: fallback plan
category={category}"]
    )
```

src/tools/create_escalation_note.py

```
from __future__ import annotations
from typing import Dict, List
from src.models import ToolResult

def create_escalation_note(ticket_case: Dict, evidence: Dict, target_team:
str) -> ToolResult:
    classification = ticket_case.get("classification", {})
    urgency = classification.get("urgency", "P4")
    category = classification.get("category", "other")

    summary = (
        f"Escalation to {target_team} | urgency={urgency} |
category={category}\n"
        f"Ticket: {ticket_case.get('ticket_text', '')[:200]}"
    )

    what_tried: List[str] = evidence.get("what_tried", [])
    recommended_next_action = evidence.get("recommended_next_action",
"Please investigate with logs and user context.")
    risk_notes: List[str] = evidence.get("risk_notes", [])

    # Add default risk note for security
    if category == "security" and "Treat as potential incident" not in
risk_notes:
        risk_notes.append("Treat as potential security incident; preserve
evidence and follow IR process.")

    return ToolResult(
        tool="create_escalation_note",
        status="ok",
        data={
            "summary": summary,
            "what_was_tried": what_tried,
            "recommended_next_action": recommended_next_action,
            "risk_notes": risk_notes
        },
        confidence=0.8,
        audit=[f"create_escalation_note: target={target_team},
category={category}, urgency={urgency}"]
    )
```

Tool registry (used in Week 3)

src/tools/registry.py

```
from __future__ import annotations
from typing import Callable, Dict, Any
from src.models import ToolResult
from src.tools.classify_ticket import classify_ticket
from src.tools.extract_requirements import extract_requirements
from src.tools.kb_search import kb_search
from src.tools.policy_check import policy_check
from src.tools.generate_resolution_plan import generate_resolution_plan
```

```
from src.tools.create_escalation_note import create_escalation_note

ToolFn = Callable[..., ToolResult]

REGISTRY: Dict[str, ToolFn] = {
    "classify_ticket": classify_ticket,
    "extract_requirements": extract_requirements,
    "kb_search": kb_search,
    "policy_check": policy_check,
    "generate_resolution_plan": generate_resolution_plan,
    "create_escalation_note": create_escalation_note,
}

def run_tool(name: str, **kwargs) -> ToolResult:
    if name not in REGISTRY:
        return ToolResult(tool=name, status="error", error=f"Unknown tool:
{name}")
    return REGISTRY[name](**kwargs)  # type: ignore
```

---

5) Unit Tests (pytest)

tests/test_classify_ticket.py

```
from src.tools.classify_ticket import classify_ticket

def test_classify_vpn():
    r = classify_ticket("VPN not connecting. Authentication failed.")
    assert r.status == "ok"
    assert r.data["category"] == "vpn"
    assert r.data["urgency"] in ["P2", "P3", "P4"]
    assert r.confidence >= 0.7

def test_classify_security():
    r = classify_ticket("Clicked a link and entered OTP, might be
phishing.")
    assert r.status == "ok"
    assert r.data["category"] == "security"
    assert r.data["urgency"] == "P1"
    assert r.confidence >= 0.8
```

tests/test_extract_requirements.py

```
from src.tools.extract_requirements import extract_requirements

def test_extract_requirements_vpn_missing_fields():
    r = extract_requirements("vpn", "VPN auth failed",
known_context={"os":"Windows"})
    assert r.status == "ok"
    assert "error_code" in r.data["fields_needed"]
    assert len(r.data["questions"]) >= 1
```

tests/test_kb_search.py

```
from src.tools.kb_search import kb_search

def test_kb_search_vpn():
    r = kb_search("vpn authentication failed windows", top_k=2)
```

```
    assert r.status == "ok"
    assert len(r.data["results"]) >= 1
    assert r.data["results"][0]["kb_id"].startswith("KB-")
```

tests/test_policy_check.py

```
from src.tools.policy_check import policy_check

def test_policy_admin_access_requires_approval():
    r = policy_check("grant_admin_access", context={"device":"laptop"})
    assert r.status == "ok"
    assert r.data["decision"] == "requires_approval"

def test_policy_unknown_action_defaults_to_requires_approval():
    r = policy_check("some_unknown_action")
    assert r.status == "ok"
    assert r.data["decision"] == "requires_approval"
```

tests/test_generate_resolution_plan.py

```
from src.tools.generate_resolution_plan import generate_resolution_plan

def test_generate_plan_uses_kb_steps():
    ticket_case = {"classification": {"category": "vpn"}}
    kb_results = [{"kb_id":"KB-VPN-
001","steps":["Step1","Step2"],"risk_level":"low"}]
    r = generate_resolution_plan(ticket_case, kb_results)
    assert r.status == "ok"
    assert r.data["steps"] == ["Step1","Step2"]
```

tests/test_create_escalation_note.py

```
from src.tools.create_escalation_note import create_escalation_note

def test_escalation_note_security_adds_risk_note():
    ticket_case = {"ticket_text":"clicked link",
"classification":{"category":"security","urgency":"P1"}}
    evidence = {"what_tried": [], "recommended_next_action":"IR followup",
"risk_notes":[]}
    r = create_escalation_note(ticket_case, evidence, target_team="SecOps")
    assert r.status == "ok"
    assert any("security incident" in x.lower() for x in
r.data["risk_notes"])
```

---

6) How to Run Week 2 Locally

```
# from repo root
python -m venv .venv
# Windows: .venv\Scripts\activate
# macOS/Linux:
source .venv/bin/activate

pip install -r requirements.txt
pytest -q
```

Expected: all tests pass.

---

7) GitHub Sync Steps (Week 2)

1) Create Week 2 branch

```
git checkout -b week2-tools
```

2) Add files + commit

```
git add .
git commit -m "Week 2: implement tools (local), add KB/policy data, add
unit tests"
```

3) Push branch

```
git push -u origin week2-tools
```

4) Create PR → merge to main

- PR Title: `Week 2: Tools + Tests`
- PR checklist:
  - ☐ `pytest` passes
  - ☐ data files present in `data/`
  - ☐ tool registry ready

---

8) Member-wise Plan (for learners in a team of 5)

Even though you're doing it solo, show them this split:

Member 1 — Tool Engineer (Classification + Requirements)

- `src/tools/classify_ticket.py`
- `src/tools/extract_requirements.py`
- tests for those 2 files

Member 2 — Tool Engineer (KB Search)

- `data/kb_index.json` + `data/kb/*.md`
- `src/tools/kb_search.py`
- test for kb_search relevance

Member 3 — Tool Engineer (Policy Engine)

- `data/policy_rules.json`
- `src/tools/policy_check.py`
- tests (known actions + unknown action)

Member 4 — Tool Engineer (Planning + Escalation)

- `src/tools/generate_resolution_plan.py`
- `src/tools/create_escalation_note.py`
- tests for both

Member 5 — Integrator / Release Manager

- `src/tools/registry.py`
- ensures all tests pass
- prepares PR description + merges

Week 3 Goal (Definition of Done)

By end of Week 3 you will have:

- A working **single agent** that:
  1. classifies the ticket
  2. asks missing questions (if needed)
  3. retrieves KB
  4. generates a step plan
  5. runs **policy gate** and decides whether HITL/escalation needed
  6. outputs in your **required format**
- A CLI demo:
  - one-shot mode (input ticket text once)
  - interactive mode (agent asks questions; user replies)
- JSONL logs for every run
- "Replay scripts" that run your Week 1 conversation paths
- Pushed to GitHub on a Week 3 branch and merged via PR

Week 3 Repo Additions

Add:

```
agentic-it-helpdesk/
  src/
    agent/
      __init__.py
      agent.py
      validators.py
      formatters.py
      logging.py
      replay.py
  scripts/
    run_agent.py
    replay_conversations.py
  logs/
    .gitkeep
```

Also update `.gitignore` to ignore `logs/*.jsonl` (keep folder tracked).

---

1) Agent Core

`src/agent/agent.py`

```
from __future__ import annotations

import uuid
```

```
from dataclasses import asdict
from typing import Any, Dict, List, Optional, Tuple

from src.models import TicketCase, ToolResult
from src.tools.registry import run_tool
from src.agent.formatters import format_response
from src.agent.validators import validate_response_sections
from src.agent.logging import JsonlLogger


CONF_THRESHOLD = 0.70


def _new_case(ticket_text: str, ticket_id: Optional[str] = None,
user_context: Optional[Dict[str, Any]] = None) -> TicketCase:
    return TicketCase(
        case_id=f"CASE-{uuid.uuid4().hex[:8]}",
        ticket_id=ticket_id,
        ticket_text=ticket_text,
        user_context=user_context or {},
    )


def _log_tool(case: TicketCase, tr: ToolResult) -> None:
    case.audit_log.append({
        "tool": tr.tool,
        "status": tr.status,
        "confidence": tr.confidence,
        "data": tr.data,
        "audit": tr.audit,
        "error": tr.error
    })


def _policy_actions_from_plan(plan_steps: List[str], category: str) ->
List[str]:
    """
    Heuristic mapping: in Week 3 we avoid external LLM parsing.
    We detect risky intents from step strings or category.
    """
    actions: List[str] = []

    # Category-driven gates
    if category == "security":
        actions.append("security_incident")

    # Step-driven gates
    joined = " ".join(plan_steps).lower()
    if "password" in joined and ("reset" in joined or "change" in joined):
        actions.append("password_reset")
    if "admin" in joined or "elevation" in joined or "privileged" in
joined:
        actions.append("grant_admin_access")

    # Safe guidance marker
    if category == "vpn":
        actions.append("vpn_guided_fix")

    # Deduplicate
    return sorted(set(actions))
```

```
def run_once(
    ticket_text: str,
    ticket_id: Optional[str] = None,
    user_context: Optional[Dict[str, Any]] = None,
    logger: Optional[JsonlLogger] = None
) -> Tuple[TicketCase, str]:
    """
    One-shot run: if missing info exists, the agent returns
request_more_info
    with questions. For full interactive resolution, use `run_interactive`.
    """
    case = _new_case(ticket_text, ticket_id, user_context)
    logger = logger or JsonlLogger()

    # 1) CLASSIFY
    tr = run_tool("classify_ticket", ticket_text=case.ticket_text)
    _log_tool(case, tr)
    case.classification = {**tr.data, "confidence": tr.confidence}

    # 2) MISSING INFO
    if tr.confidence < CONF_THRESHOLD or
case.classification.get("category") in ("other",):
        req = run_tool(
            "extract_requirements",
            category=case.classification.get("category", "other"),
            ticket_text=case.ticket_text,
            known_context=case.user_context
        )
        _log_tool(case, req)
        case.missing_info = req.data

        response = format_response(
            case=case,
            decision="request_more_info",
            kb_used=[],
            plan=None,
            policy_flags={"hitl_required": False},
            escalation=None,
        )
        validate_response_sections(response)
        logger.log_case(case, response)
        return case, response

    # 3) KB SEARCH
    query = f"{case.classification.get('category','')} {case.ticket_text}"
    kb = run_tool("kb_search", query=query, top_k=3)
    _log_tool(case, kb)
    case.kb_results = kb.data.get("results", [])

    # 4) PLAN
    plan = run_tool("generate_resolution_plan", ticket_case=asdict(case),
kb_results=case.kb_results)
    _log_tool(case, plan)
    case.plan = plan.data

    # 5) POLICY GATE (HITL / escalation)
    category = case.classification.get("category", "other")
    actions = _policy_actions_from_plan(case.plan.get("steps", []),
category)
    policy_decisions = []
```

```
    hitl_required = False
    escalation_target = None

    for action in actions:
        pc = run_tool("policy_check", action=action, context={"category":
category, **case.user_context})
        _log_tool(case, pc)
        policy_decisions.append({"action": action, **pc.data})

        if pc.data.get("decision") in ("requires_approval",):
            hitl_required = True
        if pc.data.get("decision") in ("requires_escalation",):
            escalation_target = "SecOps"

    case.policy = {"actions": actions, "decisions": policy_decisions,
"hitl_required": hitl_required}

    # 6) Decide escalation path
    if category == "security" or escalation_target == "SecOps":
        esc = run_tool(
            "create_escalation_note",
            ticket_case=asdict(case),
            evidence={
                "what_tried": [],
                "recommended_next_action": "Start incident response triage
with preserved evidence.",
                "risk_notes": ["User reported phishing/OTP event."]
            },
            target_team="SecOps"
        )
        _log_tool(case, esc)
        case.escalation = {"needed": True, "target_team": "SecOps",
**esc.data}

        response = format_response(
            case=case,
            decision="escalate_to_SecOps",
            kb_used=case.kb_results[:1],
            plan=case.plan,
            policy_flags={"hitl_required": True},  # security -> ask
consent to file report note
            escalation=case.escalation
        )
        validate_response_sections(response)
        logger.log_case(case, response)
        return case, response

    # 7) If HITL required but not security
    if hitl_required:
        response = format_response(
            case=case,
            decision="requires_approval",
            kb_used=case.kb_results[:1],
            plan=case.plan,
            policy_flags={"hitl_required": True},
            escalation=None
        )
        validate_response_sections(response)
        logger.log_case(case, response)
        return case, response
```

```
    # 8) Normal resolution guidance
    response = format_response(
        case=case,
        decision="resolve_guidance",
        kb_used=case.kb_results[:1],
        plan=case.plan,
        policy_flags={"hitl_required": False},
        escalation=None
    )
    validate_response_sections(response)
    logger.log_case(case, response)
    return case, response


def run_interactive(
    ticket_text: str,
    ticket_id: Optional[str] = None,
    user_context: Optional[Dict[str, Any]] = None,
    logger: Optional[JsonlLogger] = None
) -> None:
    """
    Interactive run: if missing info exists, agent asks questions and user
answers.
    We update user_context and re-run until we get a non-request_more_info
decision.
    """
    logger = logger or JsonlLogger()
    context = dict(user_context or {})
    current_text = ticket_text

    while True:
        case, response = run_once(current_text, ticket_id=ticket_id,
user_context=context, logger=logger)
        print("\n" + response + "\n")

        decision = case.classification.get("decision_override")  # not
used; placeholder
        # Parse decision from response header (simple)
        if "Decision: request_more_info" in response:
            # Ask user to answer all questions in one go
            questions = case.missing_info.get("questions", [])
            if questions:
                print("Please answer these questions (one line each). Press
Enter to skip a question.\n")
                for q in questions:
                    ans = input(q + " ")
                    # naïve mapping: store answers by echoing question
index
                    if ans.strip():
                        context[f"answer_{len(context)}"] = ans.strip()
            # update text with appended answers (keeps it
deterministic/simple)
            current_text = ticket_text + "\nAdditional context: " + " |
".join([v for v in context.values() if isinstance(v, str)])
            continue

        # If HITL required, ask for approval (demo flow)
        if "Decision: requires_approval" in response or "Decision:
escalate_to_SecOps" in response:
            ans = input("Approve the required approval/escalation step?
(yes/no): ").strip().lower()
```

```
            if ans == "yes":
                print("Recorded approval. (Week 4+ could trigger a workflow
ticket; for now we stop.)")
            else:
                print("Approval not granted. (Agent stops safely.)")
            break

        # Normal resolve
        break
```

---

2) Output Formatting + Determinism

src/agent/formatters.py

```
from __future__ import annotations
from typing import Any, Dict, List, Optional

from src.models import TicketCase


def _bullet(lines: List[str]) -> str:
    return "\n".join([f"- {x}" for x in lines]) if lines else "- (none)"


def format_response(
    case: TicketCase,
    decision: str,
    kb_used: List[Dict[str, Any]],
    plan: Optional[Dict[str, Any]],
    policy_flags: Dict[str, Any],
    escalation: Optional[Dict[str, Any]],
) -> str:
    """
    Enforces the Week-1 output contract. Keep stable ordering.
    """
    category = case.classification.get("category", "other")
    subcategory = case.classification.get("subcategory", "unknown")
    urgency = case.classification.get("urgency", "P4")
    reasons = case.classification.get("reasons", [])

    kb_line = []
    for r in kb_used:
        kb_line.append(f"{r.get('kb_id')} — {r.get('title')}
(risk={r.get('risk_level')}, conf={r.get('confidence')})")

    steps = (plan or {}).get("steps", [])
    checkpoints = (plan or {}).get("checkpoints", [])
    stop_conditions = (plan or {}).get("stop_conditions", [])

    # Policy summary (human readable)
    policy_notes = []
    for d in case.policy.get("decisions", []):
        action = d.get("action")
        pdec = d.get("decision")
        approver = d.get("approver_role", "")
        policy_notes.append(f"{action}: {pdec}" + (f"
(approver={approver})" if approver else ""))
```

```
    hitl_required = bool(policy_flags.get("hitl_required", False))

    out = []
    out.append(f"## 1) Decision: {decision}")
    out.append("")
    out.append("## 2) Key Findings")
    out.append(_bullet([
        f"classification: category={category}, subcategory={subcategory}",
        f"urgency: {urgency}",
        f"reasoning signals: {', '.join(reasons) if reasons else 'n/a'}"
    ]))
    out.append("")
    out.append("## 3) Steps / Plan")
    if steps:
        out.append("\n".join([f"{i+1}. {s}" for i, s in enumerate(steps)]))
    else:
        out.append("- (no steps yet)")
    if checkpoints:
        out.append("\n**Checkpoints**\n" + _bullet(checkpoints))
    if stop_conditions:
        out.append("\n**Stop Conditions**\n" + _bullet(stop_conditions))
    out.append("")
    out.append("## 4) Policy / Safety")
    safety = [
        "I will NOT perform privileged actions (password reset/access
grants) directly.",
        "I will NOT bypass security processes or hide incidents."
    ]
    if policy_notes:
        safety.append("Policy decisions: " + "; ".join(policy_notes))
    if hitl_required:
        safety.append("HITL required: approval/confirmation needed before
next step.")
    out.append(_bullet(safety))
    out.append("")
    out.append("## 5) Evidence Used")
    out.append(_bullet([
        "user ticket text",
        f"user context keys: {', '.join(sorted(case.user_context.keys()))
if case.user_context else '(none)'}",
        "KB used: " + ("; ".join(kb_line) if kb_line else "(none)")
    ]))
    out.append("")
    out.append("## 6) Next Steps")
    next_steps = []
    if decision == "request_more_info":
        qs = case.missing_info.get("questions", [])
        next_steps.append("Please answer:")
        next_steps.extend([f"Q{i+1}: {q}" for i, q in enumerate(qs)])
    elif decision in ("requires_approval",):
        next_steps.append("This request requires approval as per policy.")
        next_steps.append("Do you want me to draft an approval request for
the approver? (yes/no)")
    elif decision in ("escalate_to_SecOps",):
        next_steps.append("This is security-sensitive. I will prepare an
escalation note to SecOps.")
        next_steps.append("Do you want me to draft and submit the incident
report note to SecOps? (yes/no)")
        if escalation:
            next_steps.append("Escalation summary preview:\n" +
escalation.get("summary", ""))
```

```
    else:
        next_steps.append("Try the steps above and tell me which step fails
(if any).")
        next_steps.append("If still unresolved, I will prepare an
escalation packet for L2.")

    out.append("\n".join(next_steps))

    return "\n".join(out)
```

---

3) Validators (output contract enforcement)

src/agent/validators.py

```
from __future__ import annotations

REQUIRED_HEADERS = [
    "## 1) Decision:",
    "## 2) Key Findings",
    "## 3) Steps / Plan",
    "## 4) Policy / Safety",
    "## 5) Evidence Used",
    "## 6) Next Steps",
]

def validate_response_sections(response: str) -> None:
    missing = [h for h in REQUIRED_HEADERS if h not in response]
    if missing:
        raise ValueError(f"Response missing required sections: {missing}")
```

---

4) JSONL Logger

src/agent/logging.py

```
from __future__ import annotations

import json
import os
from datetime import datetime
from typing import Any, Dict

from src.models import TicketCase
from dataclasses import asdict


class JsonlLogger:
    def __init__(self, log_dir: str = "logs") -> None:
        self.log_dir = log_dir
        os.makedirs(self.log_dir, exist_ok=True)
        self.path = os.path.join(self.log_dir, "runs.jsonl")

    def log_case(self, case: TicketCase, response: str) -> None:
        record: Dict[str, Any] = {
            "ts": datetime.utcnow().isoformat(),
```

```
            "case": asdict(case),
            "response": response
        }
        with open(self.path, "a", encoding="utf-8") as f:
            f.write(json.dumps(record, ensure_ascii=False) + "\n")
```

Create `logs/.gitkeep`:

```
# keep logs folder tracked; actual jsonl ignored
```

Update `.gitignore` (add):

```
logs/*.jsonl
.venv/
__pycache__/
*.pyc
```

---

5) Conversation Replay (automated demo)

src/agent/replay.py

```
from __future__ import annotations
from typing import Dict, Any, List, Optional

from src.agent.agent import run_once
from src.agent.logging import JsonlLogger


def replay_cases(cases: List[Dict[str, Any]], logger: Optional[JsonlLogger]
= None) -> List[str]:
    logger = logger or JsonlLogger()
    outputs = []
    for c in cases:
        _, response = run_once(
            ticket_text=c["ticket_text"],
            ticket_id=c.get("ticket_id"),
            user_context=c.get("user_context", {}),
            logger=logger
        )
        outputs.append(response)
    return outputs
```

---

6) CLI scripts

scripts/run_agent.py

```
from __future__ import annotations
import argparse

from src.agent.agent import run_once, run_interactive
from src.agent.logging import JsonlLogger
```

```
def main() -> None:
    parser = argparse.ArgumentParser()
    parser.add_argument("--text", type=str, required=True, help="Ticket
text")
    parser.add_argument("--interactive", action="store_true", help="Run
interactive Q/A mode")
    args = parser.parse_args()

    logger = JsonlLogger()

    if args.interactive:
        run_interactive(args.text, logger=logger)
    else:
        _, response = run_once(args.text, logger=logger)
        print(response)


if __name__ == "__main__":
    main()
```

scripts/replay_conversations.py

```
from __future__ import annotations
import json
import argparse
from src.agent.replay import replay_cases

def main() -> None:
    parser = argparse.ArgumentParser()
    parser.add_argument("--cases", type=str, required=True, help="Path to
json file with replay cases")
    args = parser.parse_args()

    with open(args.cases, "r", encoding="utf-8") as f:
        cases = json.load(f)

    outputs = replay_cases(cases)
    for i, out in enumerate(outputs, start=1):
        print("\n" + "="*80)
        print(f"REPLAY CASE {i}")
        print("="*80)
        print(out)

if __name__ == "__main__":
    main()
```

Create a replay input file:

tests/replay_cases.json

```
[
  {
    "ticket_id": "TCK-REPLAY-1",
    "ticket_text": "VPN not connecting since morning. Error: Authentication
failed.",
    "user_context": { "os": "Windows 11", "device": "Laptop" }
  },
  {
    "ticket_id": "TCK-REPLAY-2",
```

```
    "ticket_text": "I clicked a link in an email and entered OTP. Now I
think it was phishing.",
    "user_context": { "os": "Android", "device": "Mobile" }
  },
  {
    "ticket_id": "TCK-REPLAY-3",
    "ticket_text": "Need admin access to install Docker on my laptop for
project work.",
    "user_context": { "os": "macOS", "device": "Laptop" }
  }
]
```

---

7) Week 3 Tests (basic end-to-end sanity)

tests/test_agent_end_to_end.py

```
from src.agent.agent import run_once

def test_agent_vpn_resolve_guidance():
    _, resp = run_once("VPN not connecting. Authentication failed.",
user_context={"os":"Windows"})
    assert "## 1) Decision:" in resp
    assert "resolve_guidance" in resp or "request_more_info" in resp

def test_agent_security_escalates():
    _, resp = run_once("Clicked a link and entered OTP, phishing
suspected.", user_context={"device":"Mobile"})
    assert "escalate_to_SecOps" in resp
    assert "Do you want me to draft and submit the incident report note to
SecOps?" in resp

def test_agent_admin_requires_approval():
    _, resp = run_once("Need admin access to install Docker.",
user_context={"os":"macOS"})
    assert "requires_approval" in resp
```

---

8) How to Run Week 3 Locally

```
# activate venv from Week 2
source .venv/bin/activate   # macOS/Linux
# .venv\Scripts\activate    # Windows

pytest -q

# One-shot demo:
python scripts/run_agent.py --text "VPN not connecting. Authentication
failed."

# Interactive demo:
python scripts/run_agent.py --text "VPN not working" --interactive

# Replay demo (shows your 3 canonical examples):
python scripts/replay_conversations.py --cases tests/replay_cases.json
```

Outputs will also be logged to:

- `logs/runs.jsonl`

---

9) GitHub Sync Steps (Week 3)

Create Week 3 branch

```
git checkout -b week3-single-agent
```

Add + commit

```
git add .
git commit -m "Week 3: single agent loop + CLI + replay + JSONL logging"
```

Push branch

```
git push -u origin week3-single-agent
```

Open PR → merge to `main`

PR checklist:

- ☐ `pytest -q` passes
- ☐ `python scripts/replay_conversations.py --cases tests/replay_cases.json` works
- ☐ logs folder exists and `.gitignore` ignores `logs/*.jsonl`

---

10) Member-wise Plan (for learner teams of 5)

Even if you implement alone, show this split:

Member 1 — Agent Orchestrator

- `src/agent/agent.py` (state flow + tool ordering + HITL logic)

Member 2 — Output & Contract Owner

- `src/agent/formatters.py`
- `src/agent/validators.py`

- ensures output format matches Week 1 contract

Member 3 — Logging & Observability

- `src/agent/logging.py`
- ensures JSONL logs have tool audit + final response

Member 4 — Replay & Demo Engineer

- `src/agent/replay.py`
- `scripts/replay_conversations.py`
- `tests/replay_cases.json`

Member 5 — QA / Release Manager

- `tests/test_agent_end_to_end.py`
- runs tests, checks replay, prepares PR + merges

Week 4 Goal (Definition of Done)

By end of Week 4:

✅ Agent survives tool failures gracefully
✅ Logs: JSONL audit + JSONL metrics with latency per tool
✅ Config file controls thresholds, KB top_k, log paths
✅ CI pipeline in GitHub Actions runs tests on PR
✅ A "demo mode" command prints run summary + links to log file
✅ (Optional) Docker run works identically for everyone

Week 4 Repo Additions / Changes

Add:

```
agentic-it-helpdesk/
  config/
    default.yaml
  src/agent/
    metrics.py
    errors.py
    config.py
  .github/
    workflows/
      ci.yml
  Dockerfile              (optional but recommended)
  scripts/
    run_agent.py          (update: supports config + prints run_id)
  tests/
    test_agent_resilience.py
```

Update:

- `src/agent/logging.py` (add run_id + separate metrics log)
- `src/tools/registry.py` (timed tool runner + exceptions captured)
- `.gitignore` (ignore local overrides)

---

1) Config (so behavior is consistent and tunable)

`config/default.yaml`

```
agent:
  conf_threshold: 0.70
```

```
  kb_top_k: 3

logging:
  dir: logs
  audit_file: runs.jsonl
  metrics_file: metrics.jsonl

policy:
  always_escalate_security: true
  require_approval_for:
    - password_reset
    - grant_admin_access
```

src/agent/config.py

```
from __future__ import annotations
import os
import yaml
from dataclasses import dataclass
from typing import Any, Dict

@dataclass
class AgentConfig:
    conf_threshold: float = 0.70
    kb_top_k: int = 3

@dataclass
class LoggingConfig:
    dir: str = "logs"
    audit_file: str = "runs.jsonl"
    metrics_file: str = "metrics.jsonl"

@dataclass
class PolicyConfig:
    always_escalate_security: bool = True
    require_approval_for: list[str] = None  # type: ignore

@dataclass
class AppConfig:
    agent: AgentConfig
    logging: LoggingConfig
    policy: PolicyConfig

def load_config(path: str = "config/default.yaml") -> AppConfig:
    if not os.path.exists(path):
        # fallback defaults
        return AppConfig(
            agent=AgentConfig(),
            logging=LoggingConfig(),
            policy=PolicyConfig(always_escalate_security=True,
require_approval_for=["password_reset","grant_admin_access"]),
        )
    with open(path, "r", encoding="utf-8") as f:
        raw: Dict[str, Any] = yaml.safe_load(f) or {}

    agent = raw.get("agent", {})
    logging = raw.get("logging", {})
    policy = raw.get("policy", {})

    return AppConfig(
        agent=AgentConfig(
```

```
            conf_threshold=float(agent.get("conf_threshold", 0.70)),
            kb_top_k=int(agent.get("kb_top_k", 3)),
        ),
        logging=LoggingConfig(
            dir=str(logging.get("dir", "logs")),
            audit_file=str(logging.get("audit_file", "runs.jsonl")),
            metrics_file=str(logging.get("metrics_file", "metrics.jsonl")),
        ),
        policy=PolicyConfig(

always_escalate_security=bool(policy.get("always_escalate_security",
True)),
            require_approval_for=list(policy.get("require_approval_for",
["password_reset","grant_admin_access"])),
        ),
    )
```

Add `pyyaml` to `requirements.txt`:

```
pyyaml==6.0.2
```

---

2) Production-style Errors + Resilience

src/agent/errors.py

```
from __future__ import annotations

class ToolExecutionError(Exception):
    pass

class AgentValidationError(Exception):
    pass
```

---

3) Metrics + Latency Measurement

src/agent/metrics.py

```
from __future__ import annotations
from dataclasses import dataclass, asdict
from typing import Any, Dict, List
import time

@dataclass
class ToolMetric:
    tool: str
    status: str
    latency_ms: float
    confidence: float

@dataclass
class RunMetrics:
    run_id: str
    total_latency_ms: float
    decision: str
```

```
    tool_metrics: List[ToolMetric]

class Timer:
    def __init__(self):
        self.t0 = time.perf_counter()
    def stop_ms(self) -> float:
        return (time.perf_counter() - self.t0) * 1000.0

def metrics_to_dict(m: RunMetrics) -> Dict[str, Any]:
    return asdict(m)
```

---

4) Upgrade Logging: audit log + metrics log, both JSONL, with run_id

src/agent/logging.py (replace with this Week 4 version)

```
from __future__ import annotations

import json
import os
from datetime import datetime
from typing import Any, Dict, Optional

from dataclasses import asdict
from src.models import TicketCase
from src.agent.metrics import RunMetrics, metrics_to_dict
from src.agent.config import LoggingConfig


class JsonlLogger:
    def __init__(self, cfg: Optional[LoggingConfig] = None) -> None:
        cfg = cfg or LoggingConfig()
        self.log_dir = cfg.dir
        os.makedirs(self.log_dir, exist_ok=True)
        self.audit_path = os.path.join(self.log_dir, cfg.audit_file)
        self.metrics_path = os.path.join(self.log_dir, cfg.metrics_file)

    def log_case(self, run_id: str, case: TicketCase, response: str) ->
None:
        record: Dict[str, Any] = {
            "ts": datetime.utcnow().isoformat(),
            "run_id": run_id,
            "case": asdict(case),
            "response": response
        }
        with open(self.audit_path, "a", encoding="utf-8") as f:
            f.write(json.dumps(record, ensure_ascii=False) + "\n")

    def log_metrics(self, metrics: RunMetrics) -> None:
        record: Dict[str, Any] = {
            "ts": datetime.utcnow().isoformat(),
            **metrics_to_dict(metrics)
        }
        with open(self.metrics_path, "a", encoding="utf-8") as f:
            f.write(json.dumps(record, ensure_ascii=False) + "\n")
```

5) Tool Runner: time every tool call + catch exceptions

`src/tools/registry.py` (update to timed runner)

```
from __future__ import annotations
from typing import Callable, Dict, Any, Tuple
from src.models import ToolResult
from src.agent.metrics import Timer, ToolMetric

from src.tools.classify_ticket import classify_ticket
from src.tools.extract_requirements import extract_requirements
from src.tools.kb_search import kb_search
from src.tools.policy_check import policy_check
from src.tools.generate_resolution_plan import generate_resolution_plan
from src.tools.create_escalation_note import create_escalation_note

ToolFn = Callable[..., ToolResult]

REGISTRY: Dict[str, ToolFn] = {
    "classify_ticket": classify_ticket,
    "extract_requirements": extract_requirements,
    "kb_search": kb_search,
    "policy_check": policy_check,
    "generate_resolution_plan": generate_resolution_plan,
    "create_escalation_note": create_escalation_note,
}

def run_tool(name: str, **kwargs) -> ToolResult:
    if name not in REGISTRY:
        return ToolResult(tool=name, status="error", error=f"Unknown tool:
{name}")
    return REGISTRY[name](**kwargs)  # type: ignore

def run_tool_timed(name: str, **kwargs) -> Tuple[ToolResult, ToolMetric]:
    timer = Timer()
    try:
        tr = run_tool(name, **kwargs)
    except Exception as e:
        tr = ToolResult(tool=name, status="error", error=str(e),
confidence=0.0, data={})
    latency = timer.stop_ms()
    metric = ToolMetric(
        tool=name,
        status=tr.status,
        latency_ms=latency,
        confidence=float(getattr(tr, "confidence", 0.0) or 0.0),
    )
    return tr, metric
```

6) Update the Agent to use config + timed tools + metrics logs

`src/agent/agent.py` (Week 4 patch: keep your Week 3 logic, but change tool calls)

Instead of rewriting the entire file, apply these **3 changes**:

Change A: Load config + logger

At top imports, add:

```
from src.agent.config import AppConfig, load_config
from src.agent.metrics import RunMetrics
from src.tools.registry import run_tool_timed
```

Change B: In `run_once(...)`, create run_id + metrics list + total timer

Inside `run_once`, add near the beginning:

```
run_id = f"RUN-{uuid.uuid4().hex[:10]}"
tool_metrics = []
total_timer = Timer()
```

Change C: Replace each `run_tool(...)` call with `run_tool_timed(...)`

Example:

```
tr, m = run_tool_timed("classify_ticket", ticket_text=case.ticket_text)
tool_metrics.append(m)
_log_tool(case, tr)
```

Do this for every tool call.

Change D: After response is produced, compute and log metrics

Before returning:

```
total_latency = total_timer.stop_ms()
metrics = RunMetrics(
    run_id=run_id,
    total_latency_ms=total_latency,
    decision=decision_string_here,
    tool_metrics=tool_metrics
)
logger.log_case(run_id, case, response)
logger.log_metrics(metrics)
```

If you prefer, I can paste a fully merged `agent.py` in one block, but the patch approach keeps your file readable for teaching.

---

7) Improve CLI: show run_id + which log file to open

scripts/run_agent.py (update)

```
from __future__ import annotations
import argparse

from src.agent.agent import run_once, run_interactive
from src.agent.config import load_config
from src.agent.logging import JsonlLogger

def main() -> None:
    parser = argparse.ArgumentParser()
    parser.add_argument("--text", type=str, required=True, help="Ticket
text")
    parser.add_argument("--interactive", action="store_true", help="Run
interactive Q/A mode")
    parser.add_argument("--config", type=str,
default="config/default.yaml", help="Path to config yaml")
    args = parser.parse_args()

    cfg = load_config(args.config)
    logger = JsonlLogger(cfg.logging)

    if args.interactive:
        run_interactive(args.text, logger=logger)
    else:
        case, response = run_once(args.text, logger=logger)
        print(response)
        print(f"\n[Logs] audit:
{cfg.logging.dir}/{cfg.logging.audit_file}")
        print(f"[Logs] metrics:
{cfg.logging.dir}/{cfg.logging.metrics_file}")

if __name__ == "__main__":
    main()
```

---

8) Resilience Tests (simulate tool failure)

tests/test_agent_resilience.py

```
import types
from src.tools import registry
from src.agent.agent import run_once

def test_agent_survives_unknown_tool(monkeypatch):
```

```
    # Monkeypatch registry to simulate missing tool
    orig = registry.REGISTRY.get("kb_search")
    registry.REGISTRY.pop("kb_search", None)

    try:
        _, resp = run_once("VPN not connecting. Authentication failed.",
user_context={"os":"Windows"})
        # Should not crash; should still produce structured output
        assert "## 1) Decision:" in resp
    finally:
        if orig:
            registry.REGISTRY["kb_search"] = orig
```

---

9) GitHub Actions CI (pytest on every PR)

.github/workflows/ci.yml

```
name: CI

on:
  push:
    branches: [ "main" ]
  pull_request:

jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-python@v5
        with:
          python-version: "3.11"
      - name: Install dependencies
        run: |
          python -m pip install --upgrade pip
          pip install -r requirements.txt
      - name: Run tests
        run: |
          pytest -q
```

---

10) Optional but Recommended: Docker (makes learner setup easy)

Dockerfile

```
FROM python:3.11-slim

WORKDIR /app
COPY . /app

RUN pip install --no-cache-dir -r requirements.txt
```

```
CMD ["python", "scripts/run_agent.py", "--text", "VPN not connecting.
Authentication failed."]
```

Run:

```
docker build -t agentic-helpdesk .
docker run --rm agentic-helpdesk
```

---

11) How to Run Week 4 Locally

```
source .venv/bin/activate
pip install -r requirements.txt
pytest -q

python scripts/run_agent.py --text "VPN not connecting. Authentication
failed."
python scripts/run_agent.py --text "Clicked a link and entered OTP"
python scripts/run_agent.py --text "Need admin access to install Docker."
```

Then open:

- `logs/runs.jsonl`
- `logs/metrics.jsonl`

You'll see:

- each tool call latency
- total run latency
- decision outcome

---

12) GitHub Sync Steps (Week 4)

```
git checkout -b week4-mini-prod
git add .
git commit -m "Week 4: config-driven agent, metrics+latency logs,
resilience, CI pipeline"
git push -u origin week4-mini-prod
```

Open PR → merge to main.

PR checklist:

- ☐ `pytest -q` passes locally
- ☐ GitHub Actions CI passes
- ☐ logs folder tracked but `logs/*.jsonl` ignored
- ☐ metrics file created after running CLI

---

13) Member-wise Plan (how learners split Week 4 in a team of 5)

1. **Orchestrator (Member 1)**

- Patch `agent.py` to use config + timed tools + safe fallbacks

2. **Observability Engineer (Member 2)**

- `metrics.py`, logger updates, ensure latency logged per tool

3. **Platform/Config Engineer (Member 3)**

- `config/default.yaml`, `config.py`, update CLI to accept config

4. **DevOps Engineer (Member 4)**

- GitHub Actions CI + Dockerfile

5. **QA Engineer (Member 5)**

- resilience tests + runbook checks (verify logs exist and are correct)

Week 5 Goal (Definition of Done)

By end of Week 5:

✅ Agent stores and retrieves memory
✅ Agent avoids re-asking already known questions
✅ Agent avoids repeating already tried steps
✅ Memory is injected in a controlled manner ("only what's relevant")
✅ Tests cover memory read/write and "no repeat questions" behavior
✅ GitHub sync (branch + PR)

Week 5 Repo Additions / Changes

Add:

```
agentic-it-helpdesk/
  src/memory/
    __init__.py
    store.py
    schemas.py
    summarizer.py
    selectors.py
  data/
    memory.db              (created at runtime; NOT committed)
  tests/
    test_memory_store.py
    test_agent_memory_behavior.py
```

Update:

- `.gitignore` (ignore `data/memory.db`)
- `src/agent/agent.py` (inject memory into `user_context`)
- `scripts/run_agent.py` (accept `--user_id` and `--ticket_id`)

Add dependency:

- `sqlite-utils` is optional; we'll use stdlib `sqlite3` (no extra deps).

---

1) Memory Data Model (what we store)

We store two tables:

A) `user_memory` (long-term)

- `user_id`
- `key` (e.g., `os`, `device`, `vpn_client`, `preferred_language`)
- `value` (string)
- `confidence` (0–1)
- `updated_at`

B) `case_memory` (short-term)

- `case_id`
- `key` (e.g., `steps_tried`, `last_questions`, `kb_used`)
- `value` (JSON string)
- `updated_at`

---

2) Memory Store Implementation

`src/memory/store.py`

```
from __future__ import annotations
import json
import sqlite3
from dataclasses import dataclass
from datetime import datetime
from typing import Any, Dict, List, Optional, Tuple

DEFAULT_DB_PATH = "data/memory.db"

def _utc() -> str:
    return datetime.utcnow().isoformat()

@dataclass
class MemoryItem:
    key: str
    value: Any
    confidence: float = 1.0

class MemoryStore:
    def __init__(self, db_path: str = DEFAULT_DB_PATH) -> None:
        self.db_path = db_path
        self._init_db()

    def _connect(self) -> sqlite3.Connection:
        return sqlite3.connect(self.db_path)

    def _init_db(self) -> None:
        con = self._connect()
        cur = con.cursor()

        cur.execute("""
        CREATE TABLE IF NOT EXISTS user_memory (
            user_id TEXT NOT NULL,
            key TEXT NOT NULL,
            value TEXT NOT NULL,
            confidence REAL NOT NULL,
            updated_at TEXT NOT NULL,
```

```
            PRIMARY KEY (user_id, key)
        )
        """)

        cur.execute("""
        CREATE TABLE IF NOT EXISTS case_memory (
            case_id TEXT NOT NULL,
            key TEXT NOT NULL,
            value TEXT NOT NULL,
            updated_at TEXT NOT NULL,
            PRIMARY KEY (case_id, key)
        )
        """)

        con.commit()
        con.close()

    # ---------- USER MEMORY ----------
    def upsert_user_memory(self, user_id: str, item: MemoryItem) -> None:
        con = self._connect()
        cur = con.cursor()
        cur.execute("""
        INSERT INTO user_memory(user_id, key, value, confidence,
updated_at)
        VALUES (?, ?, ?, ?, ?)
        ON CONFLICT(user_id, key) DO UPDATE SET
            value=excluded.value,
            confidence=excluded.confidence,
            updated_at=excluded.updated_at
        """, (user_id, item.key, json.dumps(item.value),
float(item.confidence), _utc()))
        con.commit()
        con.close()

    def get_user_memory(self, user_id: str) -> Dict[str, Any]:
        con = self._connect()
        cur = con.cursor()
        cur.execute("SELECT key, value FROM user_memory WHERE user_id=?",
(user_id,))
        rows = cur.fetchall()
        con.close()
        return {k: json.loads(v) for (k, v) in rows}

    # ---------- CASE MEMORY ----------
    def upsert_case_memory(self, case_id: str, key: str, value: Any) ->
None:
        con = self._connect()
        cur = con.cursor()
        cur.execute("""
        INSERT INTO case_memory(case_id, key, value, updated_at)
        VALUES (?, ?, ?, ?)
        ON CONFLICT(case_id, key) DO UPDATE SET
            value=excluded.value,
            updated_at=excluded.updated_at
        """, (case_id, key, json.dumps(value), _utc()))
        con.commit()
        con.close()

    def get_case_memory(self, case_id: str) -> Dict[str, Any]:
        con = self._connect()
        cur = con.cursor()
```

```
        cur.execute("SELECT key, value FROM case_memory WHERE case_id=?",
(case_id,))
        rows = cur.fetchall()
        con.close()
        return {k: json.loads(v) for (k, v) in rows}
```

---

3) Memory Selection (MCP-lite): inject only what matters

src/memory/selectors.py

```
from __future__ import annotations
from typing import Any, Dict, List

def select_relevant_user_memory(user_mem: Dict[str, Any], category: str) ->
Dict[str, Any]:
    """
    MCP-lite: choose only relevant keys based on current ticket category.
    """
    common = ["os", "device", "preferred_language"]
    vpn = ["vpn_client"]
    access = ["manager_name"]
    security = ["mfa_enabled"]

    allow: List[str] = common.copy()
    if category == "vpn":
        allow += vpn
    if category in ("access", "software_install"):
        allow += access
    if category == "security":
        allow += security

    return {k: user_mem[k] for k in allow if k in user_mem}

def select_relevant_case_memory(case_mem: Dict[str, Any]) -> Dict[str,
Any]:
    """
    For now, case memory is small; but we still keep it controlled.
    """
    allow = ["steps_tried", "kb_used", "last_questions", "last_decision"]
    return {k: case_mem[k] for k in allow if k in case_mem}
```

---

4) Memory Summarizer (what we store automatically)

We keep it simple: we store:

- last decision
- KB used
- steps tried (from plan)

- last questions asked

src/memory/summarizer.py

```
from __future__ import annotations
from typing import Any, Dict, List

def summarize_for_case_memory(agent_response: str, plan_steps: List[str],
kb_used_ids: List[str], last_questions: List[str], decision: str) ->
Dict[str, Any]:
    return {
        "last_decision": decision,
        "kb_used": kb_used_ids,
        "steps_tried": plan_steps[:3],   # store first 3 suggested steps as
“tried” candidates
        "last_questions": last_questions[:5],
    }
```

---

5) Agent Integration Changes (Week 5)

A) Update `.gitignore`

Add:

```
data/memory.db
```

B) Update CLI to accept user/ticket IDs

`scripts/run_agent.py` (Week 5 update)

```
from __future__ import annotations
import argparse

from src.agent.agent import run_once, run_interactive
from src.agent.config import load_config
from src.agent.logging import JsonlLogger

def main() -> None:
    parser = argparse.ArgumentParser()
    parser.add_argument("--text", type=str, required=True, help="Ticket
text")
    parser.add_argument("--interactive", action="store_true")
    parser.add_argument("--config", type=str,
default="config/default.yaml")
    parser.add_argument("--user_id", type=str, default="user_demo")
    parser.add_argument("--ticket_id", type=str, default=None)
    args = parser.parse_args()

    cfg = load_config(args.config)
    logger = JsonlLogger(cfg.logging)

    if args.interactive:
        run_interactive(args.text, ticket_id=args.ticket_id,
user_id=args.user_id, logger=logger)
```

```
    else:
        _, response = run_once(args.text, ticket_id=args.ticket_id,
user_id=args.user_id, logger=logger)
        print(response)
        print(f"\n[Logs] audit:
{cfg.logging.dir}/{cfg.logging.audit_file}")
        print(f"[Logs] metrics:
{cfg.logging.dir}/{cfg.logging.metrics_file}")

if __name__ == "__main__":
    main()
```

C) Modify agent signature and inject memory

In `src/agent/agent.py`:

- add `user_id: str` parameter to `run_once` and `run_interactive`
- load memory store, fetch user+case memory, inject into context before calling tools

Below is a **drop-in patch block** you can insert near the start of `run_once` after case creation:

```
from src.memory.store import MemoryStore, MemoryItem
from src.memory.selectors import select_relevant_user_memory,
select_relevant_case_memory
from src.memory.summarizer import summarize_for_case_memory
```

Then inside `run_once` after `case = _new_case(...)`:

```
mem = MemoryStore()
user_mem_all = mem.get_user_memory(user_id)
case_mem_all = mem.get_case_memory(case.case_id)

# We don't know category yet; start with common memory only.
# After classification we will select category-specific memory.
case.user_context = {**case.user_context, **{k: user_mem_all[k] for k in
user_mem_all if k in ("os","device","preferred_language")}}

# Also inject a small slice of case memory (if rerun)
case.user_context["case_memory"] =
select_relevant_case_memory(case_mem_all)
```

After classification (once category is known), add:

```
category = case.classification.get("category", "other")
relevant_user_mem = select_relevant_user_memory(user_mem_all, category)
case.user_context = {**case.user_context, **relevant_user_mem}
```

D) After producing response, store memory

Right before returning, store:

- case memory summary
- stable user memory updates (if we learned OS/device)

Add:

```
# Persist stable user memory if present in context
if case.user_context.get("os"):
    mem.upsert_user_memory(user_id, MemoryItem(key="os",
value=case.user_context["os"], confidence=0.9))
if case.user_context.get("device"):
    mem.upsert_user_memory(user_id, MemoryItem(key="device",
value=case.user_context["device"], confidence=0.9))

# Persist case memory summary
kb_used_ids = [r.get("kb_id") for r in (case.kb_results[:1] if
case.kb_results else []) if r.get("kb_id")]
last_questions = case.missing_info.get("questions", []) if
case.missing_info else []
plan_steps = case.plan.get("steps", []) if case.plan else []

case_summary = summarize_for_case_memory(
    agent_response=response,
    plan_steps=plan_steps,
    kb_used_ids=kb_used_ids,
    last_questions=last_questions,
    decision=decision_string_here
)
mem.upsert_case_memory(case.case_id, "summary", case_summary)
mem.upsert_case_memory(case.case_id, "last_decision", decision_string_here)
```

In Week 6 we will improve this to avoid storing “steps_tried” unless user confirms they actually tried them.

---

6) Week 5 Tests

tests/test_memory_store.py

```
import os
from src.memory.store import MemoryStore, MemoryItem

def test_memory_store_user_roundtrip(tmp_path):
    db = tmp_path / "mem.db"
    ms = MemoryStore(str(db))
    ms.upsert_user_memory("u1", MemoryItem(key="os", value="Windows",
confidence=0.9))
    got = ms.get_user_memory("u1")
    assert got["os"] == "Windows"

def test_memory_store_case_roundtrip(tmp_path):
    db = tmp_path / "mem.db"
    ms = MemoryStore(str(db))
    ms.upsert_case_memory("c1", "last_decision", "resolve_guidance")
    got = ms.get_case_memory("c1")
    assert got["last_decision"] == "resolve_guidance"
```

tests/test_agent_memory_behavior.py

```
from src.agent.agent import run_once

def test_agent_uses_user_memory_to_reduce_questions(tmp_path, monkeypatch):
```

```
    # force memory db to tmp
    from src.memory import store
    store.DEFAULT_DB_PATH = str(tmp_path / "mem.db")

    # First run without os -> likely ask questions for vpn
    _, resp1 = run_once("VPN not connecting. Authentication failed.",
user_id="u1", user_context={})
    # Simulate that user had OS stored (agent would store if provided; here
we inject by second run context)
    _, resp2 = run_once("VPN not connecting. Authentication failed.",
user_id="u1", user_context={"os":"Windows 11", "device":"Laptop"})

    # We don't guarantee it always asks; but it should not ask OS question
if already known
    assert "Which operating system" not in resp2
```

---

7) How to Run Week 5

```
source .venv/bin/activate
pytest -q

# Run twice with same user_id
python scripts/run_agent.py --user_id wajahat --text "VPN not connecting.
Authentication failed."
python scripts/run_agent.py --user_id wajahat --text "VPN not connecting.
Authentication failed." --interactive
```

Then inspect `data/memory.db` locally (not committed).

---

8) GitHub Sync (Week 5)

```
git checkout -b week5-memory
git add .
git commit -m "Week 5: add SQLite memory (user + case), selective context
injection, tests"
git push -u origin week5-memory
```

PR checklist:

- ☐ `pytest -q` passes
- ☐ `data/memory.db` not committed
- ☐ agent shows fewer repeated questions for same `--user_id`

---

9) Member-wise Plan (5-member learner team)

1. **Member 1 (Memory Engineer)**

- implement `src/memory/store.py` + DB schema

2. **Member 2 (Context Control / Selector Engineer)**

- implement `selectors.py` (MCP-lite injection rules)

3. **Member 3 (Agent Integrator)**

- integrate memory into `agent.py` and CLI `--user_id`

4. **Member 4 (Summarization & Storage Rules)**

- implement `summarizer.py` + define what to store / what not

5. **Member 5 (QA)**

- memory store tests + memory behavior tests
